Saeid Samadi — Robotics Laboratory Manager

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Profile

With over six years of experience in robotics and artificial intelligence, I specialize in advancing humanoid, industrial, and bio-inspired robotic systems. My expertise includes motion planning, whole-body control, and cutting-edge robot design. Leading international collaborations and managing a state-of-the-art robotics laboratory, I have contributed to pioneering research and impactful publications. Passionate about bridging robotics and human interaction, I aim to drive innovation and create transformative solutions in the field of humanoid robotics.

Education

University of Montpellier
Ph.D. in Robotics and AI — *2018–2022*
Thesis: "Humanoid robots balance in multi-contact modes settings"

University of Tehran
M.Sc. in Mechanical Engineering — *2016–2018*
Thesis: "Designing an Online Walking Algorithm of Humanoid Robot on Rough Surfaces with Different Heights"

University of Tabriz
B.Sc. in Mechanical Engineering — *2012–2016*
Project: "Free-convection analysis of an elliptic cylinder"

Professional Experience

University of Edinburgh
Laboratory Manager (Permanent Position) — *June 2023–Present*
- Managing operations across 8 robotics laboratories
- Leading collaborations with researchers and industry partners
- Overseeing equipment maintenance and procurement
- Organizing workshops and seminars to foster innovation

University of Edinburgh
Research Associate — *Mar 2022–June 2023*
- Post-doctoral research in SLMC group
- Contributed to the HARMONY EU project on assistive robotics
- Designed control frameworks for navigation and interaction

LIRMM, Montpellier
Researcher & Ph.D. Student — *2018–Feb 2022*
- Developed real-time balance control for humanoid robots
- Conducted experiments with capacitive sensors (Fogale)
- Published works in top robotics journals and conferences

University of Tehran
Teaching Assistant — *2017–2018*
- Assisted in Smart Structures and Fuzzy Control courses
- Mentored graduate students on research projects
- Guided control system modeling and design

Skills

Programming and Development

Languages: C++, Python, MATLAB, Bash

Development Tools: GIT, Docker, CMake, YAML, Visual Studio Code, GitHub Actions

Robotics Frameworks and Tools

Frameworks: ROS (ROS1/ROS2), mc_rtc, MoveIt!

Simulation Environments: Gazebo, MuJoCo, Choreonoid, RViZ, MATLAB Simulink

Robot Platforms: Talos, HRP-4, Pepper, NAO, Quadrupeds, Robotic Arms, Wheeled robots

Hardware and Prototyping

Fabrication: CNC Machining, 3D Printing, Laser Cutting

Sensors and Actuators: Capacitive Sensors (e.g., Fogale), Force-Torque Sensors, SMA-Actuators

Control Hardware: Arduino, Raspberry Pi, Motor Drivers, Embedded Systems

Softwares and Platforms

Design Tools: SolidWorks, AutoCAD, Blender

Productivity Tools: LaTeX, Microsoft Office Suite, Google Workspace

Core Competencies

Problem-Solving: Analytical Thinking, Rapid Prototyping, System Optimization

Leadership: Team Leadership, Collaboration, Effective Communication, Project Management

Research: Scientific Writing, Data Analysis, Literature Reviews

Publications

Journal Papers

- **S. Samadi**, J. Roux, A. Tanguy, S. Caron, A. Kheddar,
  Humanoid Control Under Interchangeable Fixed and Sliding Unilateral Contacts,
  IEEE Robotics and Automation Letters (RA-L), 2021. [Link]
- C.Y. Wong, **S. Samadi**, W. Suleiman, A. Kheddar,
  Touch Semantics for Intuitive Physical Manipulation of Humanoids,
  IEEE Transactions on Human-Machine Systems, 2022. [Link]
- J. Wang, S. Kim, T.S. Lembono, W. Du, J. Shim, **S. Samadi**, et al.,
  Online Multi-Contact Receding Horizon Planning via Value Function Approximation,
  IEEE Transactions on Robotics (T-RO), 2024. [Link]

Conference Papers

- **S. Samadi**, J. Wang, H. Wang, et al.,
  NAS: N-Step Computation of All Solutions to the Footstep Planning Problem,
  IEEE-RAS International Conference on Humanoid Robots (Humanoids), 2024. [Link]
- **S. Samadi**, S. Caron, A. Tanguy, A. Kheddar,
  Balance of Humanoid Robots in a Mix of Fixed and Sliding Multi-Contact Scenarios,
  IEEE International Conference on Robotics and Automation (ICRA), 2020. [Link]
- J. Roux, **S. Samadi**, E. Kuroiwa, T. Yoshike, A. Kheddar,
  Control of Humanoids in Multiple Fixed and Moving Unilateral Contacts,
  IEEE International Conference on Advanced Robotics (ICAR), 2021. [Link]
- **S. Samadi**, A.Y. Koma, M.R. Zakerzadeh, F.N. Heravi,
  Control on SMA-Actuated Rotary Actuator by Fractional Order PID Controller,

RSI International Conference on Robotics and Mechatronics (ICRoM), 2017. [Link]

Theses

- **S. Samadi**,
  Humanoid Robots Balance in Multi-Contact Modes Settings,
  Université de Montpellier, 2021. [Link]

Under Review

- W. Du, R. Long, J. Moura, J. Wang, **S. Samadi**, S. Vijayakumar,
  An Efficient Representation of Whole-Body Model Predictive Control for Online Compliant Dual-Arm Mobile Manipulation,
  IEEE Transactions on Robotics (T-RO). [Preprint Link]

Additional Information

For a comprehensive and up-to-date list of my publications, please visit my Google Scholar Profile.

Selected Projects

Humanoid Robot Control

- **Real-Time Whole-Body Balance Control for HRP-4 (2018 - 2022)**
  Developed control frameworks for real-time balance in multi-contact scenarios on the HRP-4 humanoid robot, integrating fixed and sliding unilateral contact modes. Conducted experiments using capacitive sensors to validate the proposed algorithms.
  Supporting Work: [Paper1], [Paper2]
- **Control of Talos Humanoid Robot (2022 - Present)**
  Developed control interfaces, balance strategies, and footstep planning algorithms for Talos. Enhanced whole-body motion generation capabilities using the mc_rtc framework in SLMC group.
  Supporting Material: [Video1], [Video2], [Video3]

Motion Planning and Footstep Algorithms

- **NAS: N-Step Computation of All Solutions to the Footstep Planning Problem (2024)**
  Designed a novel N-step planning algorithm that computes all solutions to the footstep planning problem, optimizing humanoid movement over complex terrains.
  Supporting Work: [Paper]
- **Value Function Approximation for Online Multi-Contact Planning (2024)**
  Proposed a receding horizon planning approach leveraging value function approximation for real-time optimization in humanoid motion.
  Supporting Work: [Paper]

Human-Robot Interaction

- **Touch Semantics for Intuitive Physical Manipulation of Humanoids (2020 - 2022)**
  Investigated intuitive touch interactions for physical manipulation tasks with humanoids, leveraging capacitive sensor technology. Designed and tested methods enabling natural physical interactions.
  Supporting Work: [Paper]

Assistive Robotics and Healthcare Applications

- **HARMONY EU Horizon Project (2022 - 2023)**
  Contributed to enhancing healthcare through assistive robotic manipulation. Designed navigation and control strategies for wheeled robots in dynamic healthcare environments.
  Supporting Work: [Paper]
- **Autonomous Wheelchair for Healthcare Applications (Ongoing)**
  Developing an autonomous wheelchair with advanced navigation, obstacle avoidance, and user-friendly

controls to assist elderly and disabled individuals.
Under Development

AI and Smart Systems

- **Neural Fuzzy Systems and Game Strategies (2017 - 2018)**
  Designed intelligent game strategies using neural fuzzy algorithms, exploring AI techniques for complex decision-making tasks.
  Technologies: Matlab, LabView
- **Smart Structures with SMA Actuators (2016 - 2017)**
  Developed control strategies and conducted system identification experiments for SMA-actuated rotary actuators, achieving high-precision control.
  Supporting Work: [Paper]

Honors, Awards, and Services

Professional Recognition

- **Staff Award for Outstanding Contribution (2025):** Recognized for exceptional efforts as IPAB Lab Manager at the University of Edinburgh. Key contributions included bridging gaps between research projects, organizing and maintaining labs, facilitating demonstrations for visiting delegations, and fostering team spirit through social events. [Link]

Academic and Competitive Achievements

- **Reviewer for International Conferences and Journals (2018 - Present):** Journals and conferences including TRO, RAL, ICRA, IROS, and Humanoids
- **M.Sc. Fellowship by MSRT, Iran (2016):** Ranked 347th out of 15,000 in the national entrance exam
- **M.Sc. Fellowship at University of Tabriz (2016):** Among the top 10% based on undergraduate GPA
- **B.Sc. Fellowship by MSRT, Iran (2012):** Ranked in the top 0.7% out of 260,055 in the national entrance exam
- **Educational Acceleration (2007):** Completed the second grade of Middle School early due to exceptional academic performance

Online Profiles

Personal Website: saeidsamadi.github.io — *Portfolio and blog*
Google Scholar: scholar.google.com/citations — *Research publications*
GitHub: github.com/saeidsamadi — *Code repositories and open-source contributions*
LinkedIn: linkedin.com/in/saeid-samadi-6ab86ba9 — *Professional networking and career updates*
DockerHub: hub.docker.com/u/saeidsamadi — *Containerized projects*

Hobbies and Interests

Persian poetry, calligraphy, volleyball, music, traveling, reading, socializing, and exploring robotics innovations